

DAY 7

Case Studies and Final Project Presentations

FMVA Certification Prep: Financial Modeling and Valuation Bootcamp

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This chapter builds working competence in Case Studies and Final Project Presentations: the concepts, the standard procedure, and the judgment to apply it under exam conditions.

Concepts and vocabulary

Financial modeling sits at the intersection of accounting, finance, and analytical reasoning. Before you can build or critique a model, you need a shared vocabulary — the precise terms that let professionals communicate without ambiguity. This section establishes those building blocks so that every case study and final project presentation you encounter rests on a solid conceptual foundation.

Core Definitions

A financial model is a structured, quantitative representation of a business, project, or transaction, built to support a specific decision. Models translate assumptions about the real world into projected financial statements, valuations, or risk metrics.

Key vocabulary you must own:

- Assumption — an input value or relationship accepted as true for modeling purposes (e.g., revenue growth rate of 8% per year).
- Driver — a variable whose change meaningfully moves an output (e.g., unit volume is a revenue driver).
- Output — the result the model is designed to produce (e.g., Net Present Value, equity value, free cash flow).
- Scenario — a coherent set of assumptions representing one plausible future state (base, upside, downside).
- Sensitivity analysis — a test that changes one driver at a time to measure its isolated effect on an output.
- WACC (Weighted Average Cost of Capital) — the blended required return across all capital sources, used as the discount rate in DCF valuation.
- DCF (Discounted Cash Flow) — a valuation method that converts future free cash flows into present value using a discount rate.
- Model risk — the danger that a model is mis-specified, mis-used, or mis-interpreted, leading to incorrect decisions.

Where It Fits in Practice

Financial models appear across virtually every finance function: business and project valuation, M&A transaction analytics, credit decisioning, FP&A budgeting, and capital budgeting. In professional settings, models are almost universally built in Microsoft Excel, which has dominated since displacing Lotus 1-2-3 in the 1990s. Best practice separates a workbook into three distinct zones — input sheets (assumptions), calculation sheets (formulas), and output sheets (dashboards, summaries) — so that any reviewer can audit the logic without hunting through tangled cells. This transparency is not optional; it is the professional standard.

Notation

Standard notation you will see in case studies:

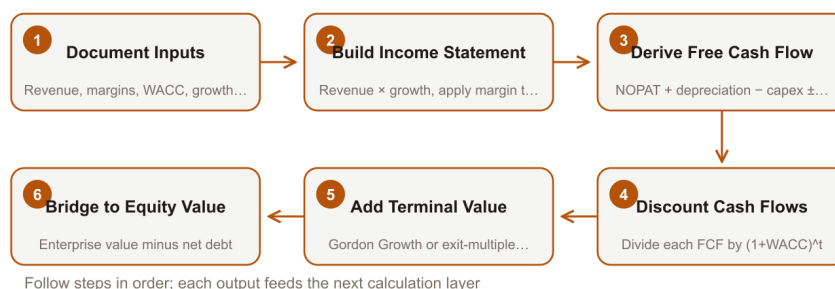
$$NPV = \sum_{t=1}^n \frac{CF_t}{(1+WACC)^t} - \text{Initial Investment}$$

$$WACC = (E/V) \times Re + (D/V) \times Rd \times (1 - \text{Tax Rate})$$

Where E = market value of equity, D = market value of debt, V = E + D, Re = cost of equity, Rd = cost of debt.

Worked Example — Terminology Applied to a Mini Scenario

DCF Model Standard Procedure



Follow steps in order; each output feeds the next calculation layer

Scenario: A small logistics firm is evaluating a new delivery route requiring a \$200,000 upfront investment. The route is expected to generate free cash flows of \$70,000 in Year 1, \$80,000 in Year 2, and \$90,000 in Year 3. The firm's WACC is 10%.

Step 1 — Identify the assumption.

The growth in cash flows (from \$70K to \$80K to \$90K) and the 10% WACC are the model's assumptions. If either changes, the output changes.

Step 2 — Identify the drivers.

Volume of deliveries and fuel cost per mile are operational drivers feeding into those cash flow figures.

Step 3 — Discount each cash flow.

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$$PV_1 = 70,000 / (1.10)^1 = \$63,636$$

$$PV_2 = 80,000 / (1.10)^2 = \$66,116$$

$$PV_3 = 90,000 / (1.10)^3 = \$67,618$$

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Step 4 — Sum the present values.

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$$\text{Total PV} = 63,636 + 66,116 + 67,618 = \$197,370$$

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Step 5 — Calculate NPV (the output).

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$$NPV = \$197,370 - \$200,000 = -\$2,630$$

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The NPV is slightly negative, suggesting the route does not meet the 10% return threshold under base-case assumptions.

Step 6 — Apply scenario thinking.

In a downside scenario (WACC rises to 12%), NPV becomes even more negative. In an upside scenario (Year 3 cash flow rises to \$100,000), NPV turns positive. This is scenario analysis in action.

Common mistake: Students often discount cash flows using the wrong period exponent — applying $(1 + \text{WACC})^1$ to Year 2 cash flows, for instance. Always match the exponent t to the time period of the cash flow, not to its position in your spreadsheet column.

Mastering these terms — assumption, driver, output, scenario, sensitivity, WACC, DCF, and model risk — gives you the language to both construct and critically evaluate any case study or final project presentation you will face in this course and in professional practice.

■ The standard procedure

Every well-built financial model follows a repeatable, auditable procedure. Mastering that procedure—inputs, steps, outputs, and checks—is what separates a model that earns stakeholder confidence from one that quietly propagates errors into decisions.

■ Inputs

Before a single formula is written, gather and document every assumption explicitly. Best practice separates inputs onto a dedicated sheet so that any reviewer can trace every number back to its source without hunting through calculation layers.

For a discounted cash flow (DCF) valuation, the standard inputs are:

- Revenue base and growth rates (historical actuals plus forward assumptions)
- Operating margin (EBIT or EBITDA margin, sourced from management guidance or industry benchmarks)
- Tax rate (effective, not statutory, unless you have a reason to use statutory)
- Capital expenditure and depreciation schedules
- Working capital assumptions (days sales outstanding, days payable outstanding, inventory days)
- Discount rate — typically the Weighted Average Cost of Capital (WACC)
- Terminal growth rate

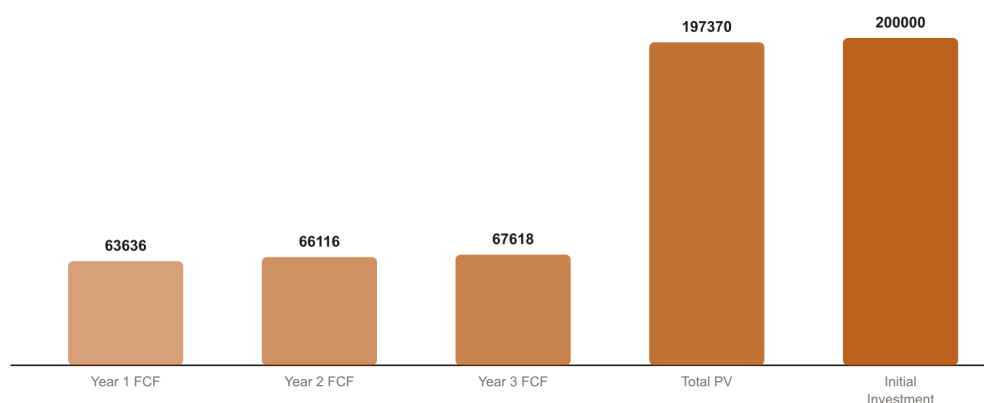
Label every input cell, include units, and note the source. This is not optional housekeeping; it is the foundation of model auditability.

Steps

- Build the income statement projection. Start with revenue, apply the growth rate, then layer in costs using the margin assumption to arrive at EBIT.
- Calculate taxes and NOPAT. Apply the effective tax rate to EBIT to get Net Operating Profit After Tax.
- Derive free cash flow (FCF). Add back non-cash charges (depreciation), subtract capital expenditure, and adjust for changes in net working capital.
- Discount each FCF to present value. Apply the WACC as the discount rate across the forecast horizon.
- Estimate terminal value. Use the Gordon Growth Model or an exit-multiple approach.
- Sum to enterprise value. Add the PV of FCFs to the PV of terminal value.
- Bridge to equity value. Subtract net debt (debt minus cash) from enterprise value.

Worked Numerical Example

DCF Example: Cash Flow Present Values



Total PV \$197,370 minus \$200,000 investment yields NPV of -\$2,630

Total PV \$197,370 minus \$200,000 investment yields NPV of " \$2,630

Scenario: A single-year simplified DCF for a manufacturing firm.

| Input | Value |

|---|---|

| Current revenue | \$10,000,000 |

| Revenue growth rate | 8% |

| EBIT margin | 20% |

| Effective tax rate | 25% |

| Depreciation | \$300,000 |

| Capital expenditure | \$500,000 |

| Change in net working capital | \$150,000 |

| WACC | 10% |

| Terminal growth rate | 3% |

| Net debt | \$2,000,000 |

Step 1 — Projected revenue:

$$\text{Revenue} = \$10,000,000 \times 1.08 = \$10,800,000$$

Step 2 — EBIT:

$$\text{EBIT} = \$10,800,000 \times 0.20 = \$2,160,000$$

Step 3 — NOPAT:

$$\text{NOPAT} = \$2,160,000 \times (1 - 0.25) = \$1,620,000$$

Step 4 — Free cash flow:

$$\text{FCF} = \text{NOPAT} + \text{Depreciation} - \text{CapEx} - \Delta \text{NWC}$$

$$\text{FCF} = \$1,620,000 + \$300,000 - \$500,000 - \$150,000 = \$1,270,000$$

Step 5 — PV of Year 1 FCF:

$$\text{PV} = \$1,270,000 \div 1.10 = \$1,154,545$$

Step 6 — Terminal value (Gordon Growth Model):

$$\text{TV} = \text{FCF} \times (1 + g) \div (\text{WACC} - g) = \$1,270,000 \times 1.03 \div (0.10 - 0.03) = \$18,690,000$$

Step 7 — PV of terminal value:

$$\text{PV of TV} = \$18,690,000 \div 1.10 = \$16,990,909$$

Step 8 — Enterprise value:

$$EV = \$1,154,545 + \$16,990,909 = \$18,145,454$$

Step 9 — Equity value:

$$\text{Equity Value} = \$18,145,454 - \$2,000,000 = \$16,145,454$$

Outputs

The primary output is the equity value, which can be divided by shares outstanding to yield a per-share intrinsic value. Secondary outputs include a sensitivity table (WACC vs. terminal growth rate) and a scenario summary (base, bull, bear cases). Always present outputs on a separate sheet from calculations.

Checks

Run these before sharing any model:

- Sum-of-parts check: Confirm that FCF components reconcile to the income statement and balance sheet.
- Circular reference audit: Flag any circular logic, particularly in interest expense calculations.
- Sensitivity sanity check: Verify that equity value moves in the expected direction when WACC rises (it should fall).
- Peer benchmarking: Compare your implied EV/EBITDA multiple to industry comparables. A wildly different multiple signals a flawed assumption.

Common mistake: Analysts frequently forget to discount the terminal value back to the present. The Gordon Growth Model produces a value at the end of the forecast period, not today. Dividing by $(1 + WACC)^n$ is mandatory, and omitting this step inflates equity value substantially.

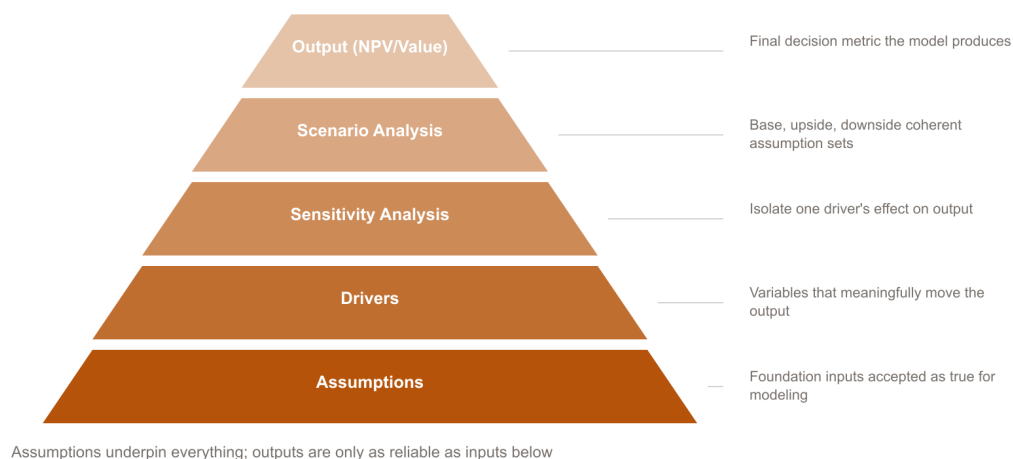
Disciplined adherence to this procedure—document inputs, execute steps sequentially, present clean outputs, and run systematic checks—is what makes a financial model both defensible and useful under scrutiny.

Applications and pitfalls

Financial models serve a remarkably broad range of professional purposes, from valuing a startup in a venture capital pitch to stress-testing a municipal infrastructure project under multiple economic scenarios. Understanding where these tools shine—and where practitioners routinely stumble—is what separates a competent analyst from a dangerous one.

Typical Use Cases

Financial Model Vocabulary Hierarchy



Assumptions underpin everything; outputs are only as reliable as inputs below

The most common applications cluster around three activities: valuation, planning, and decision support. Discounted cash flow (DCF) models underpin business, stock, and project valuation. Scenario and sensitivity models drive FP&A cycles and strategic planning ("what if revenue drops 20%?"). Capital budgeting models evaluate whether a new factory or acquisition clears the hurdle rate. Credit models assess borrower default risk, while transaction models—LBOs, M&A, project finance—layer in deal-specific mechanics like debt waterfalls and earnouts.

Across all these contexts, the model is only as trustworthy as its structure. Professional best practice calls for separating input sheets (assumptions), calculation sheets (logic), and output sheets (results). This separation makes the model auditable, reduces formula errors, and builds stakeholder confidence. Consistent formatting, labeled ranges, and documented assumptions are not cosmetic niceties—they are risk controls.

Common Mistakes

Practitioners and researchers have catalogued "spreadsheet risk" extensively. The most damaging errors tend to be:

- Hard-coding numbers inside formulas instead of referencing a dedicated assumptions cell
- Circular references introduced accidentally when linking statements
- Inconsistent time periods—mixing annual and monthly figures in the same row
- Omitting the terminal value in a DCF or double-counting it
- Ignoring working capital changes when converting net income to free cash flow
- Overfitting scenarios to a desired answer rather than stress-testing honestly

Model risk—the danger that a technically correct-looking model produces incorrect pricing or decisions—is a growing area of risk management research precisely because these errors can cascade silently.

Sanity Checks

Before presenting any model, run these checks:

- Sum-of-parts test — Do individual line items add up to the stated totals?

- Balance sheet balances — Assets must equal liabilities plus equity in every period.
- Cash flow reconciliation — Beginning cash + net cash flows = ending cash.
- Ratio plausibility — Are margins, coverage ratios, and growth rates within industry norms?
- Extreme-case test — Set revenue to zero; does the model break or produce a sensible loss?
- Independent recalculation — Manually verify at least one period end-to-end.

Worked Example: A Flawed DCF, Corrected

Scenario: An analyst values a small manufacturing firm. Projected free cash flow (FCF) for Year 1 is \$500,000, growing at 4% per year in perpetuity. The firm's WACC is 9%. The analyst calculates terminal value (TV) and enterprise value (EV) as follows.

The Flawed Attempt

The analyst uses the Gordon Growth Model for terminal value but mistakenly applies it to Year 0 cash flow (the current year's actual FCF of \$480,000) and forgets to discount the TV back to today.

$$TV = FCF_0 / (WACC - g) = 480,000 / (0.09 - 0.04) = \$9,600,000$$

$$EV = FCF_0 + TV = 500,000 + 9,600,000 = \$10,100,000 \quad \text{!• WRONG}$$

Two errors: wrong FCF in the numerator, and TV is not discounted.

Step-by-Step Correction

Step 1 — Identify the correct cash flow for the terminal value formula.

The Gordon Growth Model requires the next period's cash flow beyond the explicit forecast horizon. If Year 1 FCF is \$500,000 and the terminal value is calculated at the end of Year 1, the numerator is Year 2 FCF:

$$FCF_2 = FCF_1 \times (1 + g) = 500,000 \times 1.04 = \$520,000$$

Step 2 — Calculate terminal value at end of Year 1.

$$TV_1 = FCF_2 / (WACC - g) = 520,000 / (0.09 - 0.04) = \$10,400,000$$

Step 3 — Discount all cash flows to today (Year 0).

$$PV \text{ of } FCF_1 = 500,000 / 1.09 = \$458,716$$

$$PV \text{ of } TV_1 = 10,400,000 / 1.09 = \$9,541,284$$

Step 4 — Sum to enterprise value.

$$EV = 458,716 + 9,541,284 = \$10,000,000$$

The corrected EV is \$10,000,000—roughly \$100,000 lower than the flawed figure, a difference that could meaningfully affect a purchase price negotiation.

Common mistake: Analysts frequently forget that the terminal value sits at the end of the forecast period and must be discounted back like any other cash flow. Leaving TV undiscounted inflates enterprise value and can make a marginal deal appear attractive.

Final Plausibility Check

Divide EV by Year 1 FCF: $\$10,000,000 / \$500,000 = 20\times$. For a stable manufacturing business with 4% growth, an FCF multiple of $20\times$ is reasonable (it equals $1 / (\text{WACC} - g) = 1 / 0.05$). The model passes the sanity check.

Disciplined application of these steps—correct inputs, proper discounting, and an independent plausibility review—is what transforms a spreadsheet into a reliable decision-making tool.

